EE-C-002143(Army-GL)
June 27, 1983

INTERIM FEDERAL SPECIFICATION

CAKE, COFFEE, APPLE, TRAY PACK

This interim Federal specification was developed by the US Army Natick Research and Development Laboratories, Natick, MA 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers coffee cake in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

EE-S-631	-	Spices, Ground and Whole, and Spice Blends
TT-C-495	-	Coatings, Exterior for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification, and other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

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(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-35078	-	Loads, Unit: Preparation of Nonperishable Subsistence, Items General Specification for

Military Standard:

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-900	-	Bacterial Standards for Starches, Flour, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Canned Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

US Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

US Department of Agriculture (USDA)

Regulations Governing the Grading and Inspection of Eggs and Egg Products (7 CFR Part 59)

US Standards for Grades of Nonfat Dry Milk (Spray Dried)

(Application for copies should be addressed to the Director, Dairy Division, Agricultural Marketing Service, US Department of Agriculture, Washington, DC 20250.)

US Standards for Grades of Fruit Jelly

(Application for copies should be addressed to the Chief, Processed Products Branch, Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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US Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Food Quality Assurance Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

American Oil Chemists Society

Official and Tentative Methods of the American Oil Chemists Society

(Application for copies should be addressed to the American Oil Chemists Society, 508 South Sixth Street, Champaign, IL 61820.)

National Academy of Sciences

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, Washington, DC 20418.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1 and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

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3.2.1 Apples. Apples shall be dehydrated, diced, non-sweetened with less than 5.0 percent moisture.

3.2.2 Apple jelly. Apple jelly shall be as per FDA Standards of Identity and US Grade B or better.

3.2.3 Chemical leavening agents. Chemical leavening agents shall be bicarbonate of soda, and sodium aluminum phosphate. The coating shall be a hard fat or other suitable material such as to slow the solubility of the salt. The above ingredients shall be powdered and free flowing.

3.2.4 Cinnamon. The cinnamon shall be type II of EE-S-631.

3.2.5 Egg products. Egg products shall be fresh whole eggs (with the formula water adjusted) or dried whole eggs, produced and handled in compliance with the requirements set forth in the US Department of Agriculture publication entitled "Regulations Governing the Inspection of Eggs and Egg Products". They shall have been prepared under continuous inspection of USDA and shall bear the inspection shield mark.

3.2.6 Flour. Flour shall be hard or soft wheat of the type known as high ratio cake flour and enriched. If unenriched flour is used, equivalent enrichment shall be incorporated by the manufacturer at the time of production of the finished product. In addition, flour shall comply with MIL-STD-900.

3.2.7 Glycerol. Glycerol shall be Food Grade and shall meet the requirements of the Food Chemicals Codex.

3.2.8 Mace. The mace shall be type II and shall be derived from the dried arillus of Myristica fragrans Houtt. The ground mace shall be yellowish-tan to reddish-tan in color and shall have fragrant, nutmeg-like, aromatic odor, a slightly warm taste and shall be uniformly ground, conforming to EE-S-631.

3.2.9 Mono-diglycerides. Mono-diglycerides shall be a combination of monoglycerides and diglycerides with manufacturer's typical analyses indicating the alpha monoglyceride content to be between 40 and 55 percent.

3.2.10 Nonfat dry milk. Nonfat dry milk shall be of US Extra grade as defined in the US Standards for Grades of Nonfat Dry Milk. It shall be of the high heat type.

3.2.11 Nutmeg. The nutmeg shall be type II and shall be derived from dried seeds of Myristica fragrans Houtt. The ground nutmeg shall have a dull brown color and shall be uniformly ground, conforming to EE-S-631. Volatile oil content shall be not less than 1.0 milliliters per 100 grams.

3.2.12 Salt. Salt shall be white refined sodium chloride with or without anticaking agent. Iodized salt shall not be used.

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3.2.13 Shortening. Shortening shall be refined, hydrogenated vegetable oil having a stability of not less than 100 hours as determined by the Active Oxygen Method (AOM) in Method: Cd 12-57, Chapter: Commercial Fats and Oils, of the "Official and Tentative Methods of American Oil Chemists Society". The shortening shall contain antioxidants with active ingredients of BHQ, BHA, BHT, propyl gallate, alone or in combination in accordance with good commercial practice at the maximum allowable level permitted by the FDA.

3.2.14 Sodium stearoyl-2-lactylate. Sodium stearoyl-2-lactylate shall be powdered and free flowing and comply with the requirements of Food Chemicals Codex.

3.2.5 Sugar. Sugar shall be white, refined cane or beet sugar, or combination thereof.

3.3 Formulation. The formulation for the batter and spice mix coating shall be as follows:

Ingredients	Parts by weight
Batter	
Sugar	27.50
Flour	26 80
Water	17.40
Egg products	10.80
Apple jelly	6.40
Shortening	5.80
Glycerol	1.50
Non fat dried milk	1.40
Salt	0.60
Mono-diglycerides	0.60
Sodium aluminum phosphate	0.40
Bicarbonate of soda	0.40
Sodium stearoyl-2-lactylate	0.30
Mace	0.10
FD & C yellow #5	*
	100.00

*NOTE: 16.75 grams FD & C yellow #5 per 1000 pounds of dry mix.

Coating	
Apples	68.0
Sugar	16.9
Water	14.0
Cinnamon	1.0
Nutmeg	0.1
	100.00

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3.4 Batter preparation and tray pack filling, processing, and sealing.

a. Premix all of the dry base ingredients for 5 minutes at low speed.
b. Mix water, glycerol, and jelly together.
c. Pour 1/2 of water mixture into the dry mix.
d. Mix 1 minute at low speed.
e. Scrape down the bowl.
f. Add remainder of water mixture to the batter.
g. Mix 1 minute at low speed.
h. Mix 2 minutes at medium speed.
i. Scrape down bowl.
j. Mix for 4 minutes at medium speed.
k. Place a 28.5 cm by 22 cm piece of parchment with rounded corners in the bottom of the tray pack. 1/
l. Fill tray with batter (48 ounces) and level it off.
m. To the diced apples add the water and mix till absorbed, then mix in spice and sugar mix.
n. Place 5 ounces (per tray) diced coated apples over the surface.
o. Place a 31 cm by 24.9 cm piece of parchment paper with rounded corners over the coated batter. 1/
p. Place a lid over the tray pack.
q. Place a 5 to 7 pound weight over the lid or crimp cover down.
r. Place the tray pack in a 275°F rotary oven for 90 minutes, until completely baked in the center.
s. Remove the covered tray pack from the oven and place in a high efficiency air filter chamber to cool to 170°F (internal cake temperature). This takes approximately 15 minutes at 68°F or 30 minutes at 80°F, ambient temperature.
t. Seal trays with 4 inches of vacuum. Do not allow buckling of the corners.

1/ The parchment paper shall have a minimum basis weight of 27 pounds per ream (24 by 36 - 500).

3.5 Finished product. The cake shall be evenly baked with no raw, stringy, or ungelatinized portions, and shall be lightly browned, sugar crusted with visible apples on top. When the baked coffee cake is bisected vertically with a sharp knife, it shall not crumble nor show any compression streaks, gummy interior or soggy areas, nor be excessively dry or overbaked and the interior crumb shall have a close even grain. The texture shall be soft and smooth and the crumb color shall be yellow with yellow-beige apples and brown spice. The product shall have a sweet, yellow cake and apple flavor, and shall be free from rancid, scorched or other objectionable flavors and odors. The water activity of the coffee cake shall be below 0.80. The average net weight of the filled and sealed cans shall be a minimum of 52.0 ounces. No individual can shall contain less than 51.0 ounces of product.

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3.6 Applesauce. Applesauce shall be canned sweetened applesauce of U.S. Grade

3.5.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance.

3.6 Plant qualification. The product shall be prepared, processed and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable government inspection ageny.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, US Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any non-conformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

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4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans, liners, and lids used in preparing the product and dividing by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 0.1 ounce. If the average net weight is less than 52.0 ounces or if the net weight of any individual can is less than 51.0 ounces, the lot shall be rejected.

4.6.2 Product inspection. Eight tray packs shall be selected at random from the lot. The tray packs shall be opened and inspected for the defects listed in table I. Finding of any major defect shall be cause for rejection of the lot.

TABLE I. Product inspection 1/ 2/

Category	Defect
Major	
101	Crust not an even golden brown color.
102	When bisected vertically with a sharp knife, the cake crumbles; has compression streaks, a gummy center, or soggy areas; does not have a reasonably close, even, grain structure; has raw, stringy, or ungelatinized portions; or is excessively over-baked.
103	Does not have a pleasant yellow cake and apple-cinnamon flavor.

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TABLE I. Product inspection 1/ 2/(cont'd)

Category	Defect
Major	
104	Texture not soft and smooth.
105	Inedible, hard or mushy apple pieces.
106	Non-uniform distribution of apple pieces visible on the surface.
107	Crumb color not yellow, apple pieces not yellow-beige, or spice not brown.
108	Distinct apple pieces not present.

1/ The presence of any foreign material (e.g., glass, dirt, insect parts, hair, wood, metal, etc.) foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale, etc.) or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

4.6.3 Water activity testing. Five tray packs shall be selected at random from the lot and tested for water activity in accordance with Method 32.004 in the Vegetable Products chapter of the Official Methods of Analysis of the Association of Official Analytical Chemists. Two determinations shall be made on each tray pack cake in the sample: one on a specimen from the core in the center of the cake and the other on a specimen from the outside edge. The results shall be reported to the nearest 0.01 percent. If the average water activity of any cake in the sample is 0.80 or above, the lot shall be rejected.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Reinforcement with nonmetallic strapping or tape is not used.

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4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with food product, baked, and sealed as specified in 3.4 to achieve a net weight of not less than 52.0 avoirdupois ounces. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient weight and temper to protect the product during shipment and storage. The entire inside area of the can shall be coated. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pads shall be full length and width less approximately 1/4 inch in each dimension. The box liner and pads shall be fabricated of grade V3c fiberboard. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The pads shall be placed between the cans and on top and bottom of the stacked cans. The pads shall be full length and width less approximately 1/4 inch in each dimension and be fabricated of class domestic fiberboard.

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5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray can shall be labeled in accordance with MIL-L-1497.

Paper labels are not permitted. Cans shall show the following statements:

DO NOT HEAT

Yield

Approximately 20 servings per tray cut 4 slices in width by 5 slices in length.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.1, 4.3 and 6.3).
c. Provisions for approved preproduction samples (see 3.5.1 and 6.3).
d. Level of packaging and packing required (see 5.1 and 5.2).
e. Type and class of unit load when loading is required (see 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

Preparing activity:

Army - GL

Project No. 8920-A431

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
EE-C-002143(Army-GL)	Cake, Coffee, Apple, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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of one pouch face. Hydrated dehydrated